

THINKING

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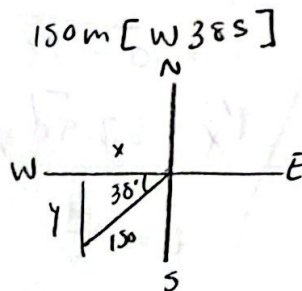
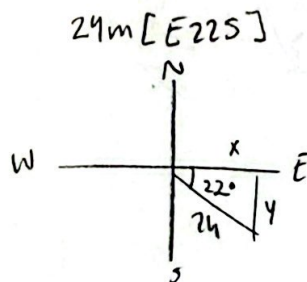
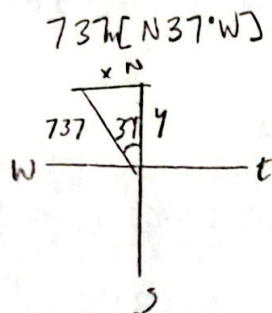
APPLICATION

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Answer the following questions CLEARLY showing all your work. Isolate for the unknown variable, units, directions.... Round your final answer to 3 sd and angle to the nearest number.

THINKING – (10 marks)

1. Dave walked all the way from the bus stop to class covering a total of 737 m [N 37° W]. Mr. Tacchino sends him away because he is still wearing a sweater. Dave walks 24 m [E 22° S] and then another 150 m [W 38° S] to put his sweater in his locker until the end of the day. Use algebra to solve for Dave's final displacement. (4 marks)



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$$d_{1x} = 737 \sin 37^\circ \text{ m [W]}$$

$$= 443.59 \text{ m [W]}$$

$$d_{1y} = 737 \cos 37^\circ \text{ m [N]}$$

$$= 588.59 \text{ m [N]}$$

$$= -588.59 \text{ m [S]}$$

$$d_{Tx} = 443.59 \text{ m [W]} + (-22.852 \text{ m [W]}) + 118.20 \text{ m [W]}$$

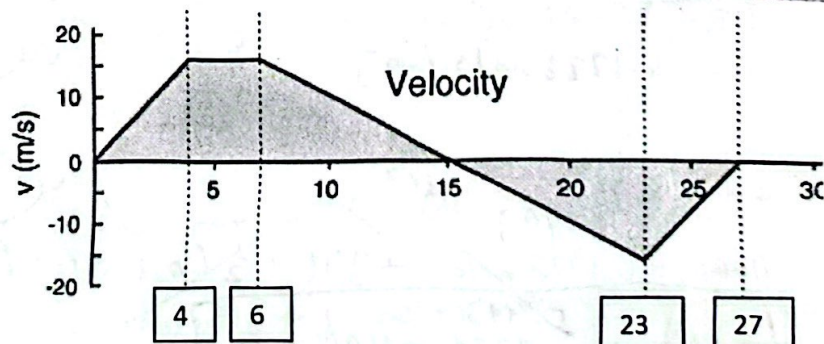
$$= 538.88 \text{ m [W]}$$

$$d_{Ty} = -588.59 \text{ m [S]} + 8.99 \text{ m [S]} + 92.399 \text{ m [S]}$$

$$= -487.25 \text{ m [S]} \rightarrow 487.25 \text{ m [N]}$$

$$d_T = \sqrt{487.25^2 + 538.88^2}$$

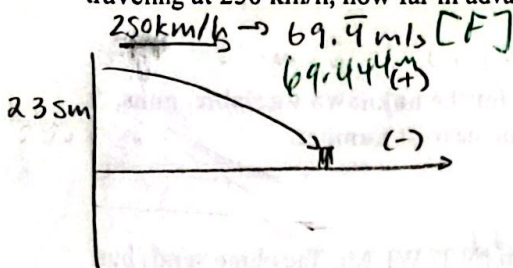
2. A velocity time graph depicts the motion for an object; take positive direction to be **NORTH** and time to **SECONDS**.



- a. State the time(s) when is the object not moving. $t = 0, 15, 27$ (1 mark)
- b. State the time(s) when the object is moving at a constant velocity $t = 4\text{s} - 6\text{s}$ (1 mark)

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3. A rescue plane wants to drop supplies to isolated mountain climbers on a rocky ridge 235 m below. If the plane is traveling at 250 km/h, how far in advance of the climbers (horizontal distance) must the packages be dropped? (4 marks)



$$-235 = 0 - \frac{1}{2}(-9.81)t^2$$

$$t = 6.95$$

$$d = 6.95(69.4) = 0.0999$$

$$d = -235 \text{ m}$$

$$v_y = 0 \text{ m/s}$$

$$a = -9.81 \text{ m/s}^2$$

$$t = ?$$

$$v_y = ?$$

$$v_y^2 = v_{iy}^2 + 2ad$$

$$v_1 = \sqrt{v^2 - 2ad}$$

$$v_1 = \sqrt{0 - 2(-235)(-9.81)}$$

$$v_x = 67.93 \text{ m/s}$$

$$t = \frac{d}{v}$$

$$t = \frac{235 \text{ m}}{67.9 \text{ m/s}}$$

$$= 3.46 \text{ m away}$$

it must be dropped 3.46 m away

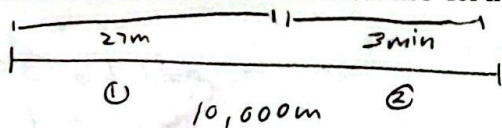
$$[0.0997 \text{ m/s} \rightarrow]$$

$$(1) t = 6.9252 \text{ s}$$

$$(2) d_x = v_x \cdot t = 48 \text{ m before}$$

APPLICATION - (10 marks)

1. A runner hopes to complete the 10 000 m run in less than 30 min. After 27 min, there is still 1100 m to go. The runner must then accelerate at 0.2 m/s^2 for how many seconds in order to achieve the desired time? (4 marks)



after 27 min, 8900 m is traveled

$$(2) d = 1100 \text{ m} \rightarrow$$

$$a = 0.2 \text{ m/s}^2 \rightarrow$$

$$t = ?$$

$$v_i = 6.1728 \text{ m/s} \rightarrow$$

$$(1) v = \frac{d}{t}$$

$$d = 10,000 \text{ m} \rightarrow$$

$$t = 27 \text{ min} \rightarrow 1620 \text{ sec}$$

$$v = \frac{10,000 \text{ m}}{1620 \text{ s}} = 5.49 \text{ m/s}$$

$$v = 6.1728 \text{ m/s} \rightarrow$$

$$\Delta d = v_i \Delta t + \frac{1}{2} a \Delta t^2$$

$$1100 \text{ m} = (6.1728 \text{ m/s})(t) + \frac{1}{2}(0.2 \text{ m/s}^2)t^2$$

$$0.1t^2 + 6.1728t - 1100 = 0$$

$$-6.1728 \pm \sqrt{6.1728^2 - 4(0.1)(-1100)}$$

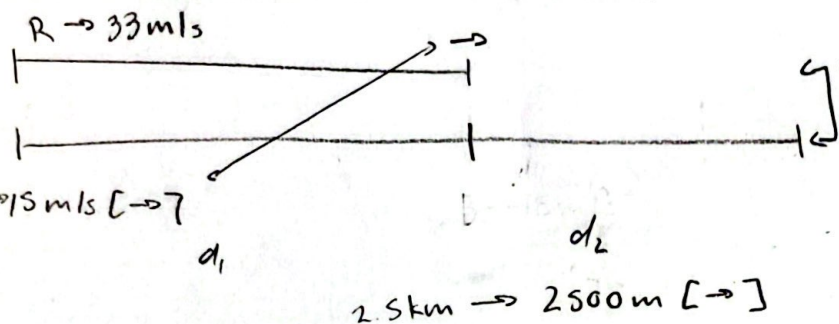
$$t = 78.46 \text{ s} - 140.19 \text{ s} = 80.95 \text{ s}$$

it will take 78.5 seconds for the runner to achieve the desired time

A red car is traveling at a constant velocity of 33 m/s on a highway. At the instant the red car passes the end of an entrance ramp, a blue car enters the highway from the ramp. The blue car coast along the ramp at 15 m/s, and when it reaches the end of the entrance ramp, the blue car undergoes a constant acceleration.



What acceleration must the blue car maintain and what will be its velocity when both cars meet for the first time at the next exit 2.5 km away? (6 marks)



$$R: v = 33 \text{ m/s } [\rightarrow]$$

$$B: v_1 = 15 \text{ m/s}$$

$$d = 2500 \text{ m } [\rightarrow]$$

$$a = ?$$

$$d = 2500 \text{ m } [\rightarrow]$$

$$v_2 = ?$$

$$t = t$$

$$R: t = \frac{d}{v}$$

unit

$$B: d = v_1 t + \frac{1}{2} a t^2$$

$$t = \frac{2500 \text{ m}}{33 \text{ m/s}}$$

$$t = 75.7575 \text{ s}$$

$$a = \frac{d - v_1 t}{\frac{1}{2} t^2}$$

$$a = \frac{2500 \text{ m} - (15 \text{ m/s})(75.7575 \text{ s})}{\frac{1}{2} (75.7575 \text{ s})^2}$$

$$a = 0.47521 \text{ m/s}^2 [\rightarrow]$$

$$v_2 = v_1 + a \Delta t$$

$$v_2 = 15 \text{ m/s} + (0.47521 \text{ m/s}^2)(75.7575 \text{ s})$$

$$v_2 = 15.228 \text{ m/s } [\rightarrow]$$

$$2500 = d_1 + d_2$$

$$d_2 = 2500 - d_1$$

the blue car must maintain
a $0.475 \text{ m/s}^2 [\rightarrow]$ acceleration
and its final velocity will
be $15.2 \text{ m/s } [\rightarrow]$ $51 \text{ m/s } [\rightarrow]$